

Systematic Map Protocol

Title

Mapping direct anthropogenic drivers of biodiversity change using large language models: a systematic map protocol

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Background

Achieving the 2050 Vision of the Kunming-Montreal Global Biodiversity Framework requires understanding what drives biodiversity loss across all drivers, ecosystems, and geographies [1]. Direct anthropogenic drivers act immediately on ecosystems, species, and populations [1,2], yet the scientific evidence base on these drivers remains uneven. A small subset of taxa receives the majority of research attention [3], and the same imbalance holds across drivers: climate change accounts for nearly 40% of driver-related publications, while pollution is represented in fewer than 5% [4]. Most evidence syntheses address a single driver, taxon, or region [5–9], leaving the broader evidence base across drivers largely uncharacterized. This systematic map will apply a broad search strategy for biodiversity change and direct anthropogenic drivers to screen over 2.4 million candidate bibliographic records. Following the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) conceptual framework [2], eligible records will be coded to produce a structured evidence base across drivers, threats, geography, ecosystems, study attributes, and taxa. The rapidly expanding biodiversity literature makes comprehensive manual synthesis increasingly challenging [10]; therefore, this map will use Large Language Models (LLMs) to automate screening and coding within the Collaboration for Environmental Evidence (CEE) systematic mapping framework. The map will help biodiversity researchers and policymakers identify evidence gaps and research priorities across direct anthropogenic drivers.

Theory of change or causal model

This systematic map does not evaluate interventions and does not construct a causal model. Following the IPBES conceptual framework [2], it characterizes the scientific evidence base on direct anthropogenic drivers of biodiversity change. The map draws on this framework to define the exposure (direct anthropogenic drivers) and outcome (biodiversity change), as illustrated in Supplementary Figure 1.

Stakeholder engagement

No formal external stakeholder consultation will be conducted. However, screening and coding outputs will be compared against labels from previously published sources on biodiversity change through the consistency checking procedures described in this protocol.

Objectives and review question

The objective of this systematic map is to collect, code, and characterize the scientific evidence on the direct anthropogenic drivers of biodiversity change, with primary analytical focus on biodiversity loss. The map is guided by two overarching questions: 1. What scientific evidence is available on the direct anthropogenic drivers of biodiversity change? 2. How is this evidence characterized across direct anthropogenic drivers, specific threats, geography, ecosystems, study attributes, and taxa?

Definitions of the question components

The review questions are defined using the Population, Exposure, and Outcome (PEO) framework. Population: This map broadly covers total biodiversity, the variability among living organisms from all sources [11]. No geographic or taxonomic restriction is applied; all taxonomic groups, ecosystems, and geographies will be coded using pre-defined classification schemes [2, 12–15]. Exposure: The exposure comprises any direct anthropogenic driver acting on biodiversity, classified using the five IPBES driver categories [2] and further resolved using the International Union for Conservation of Nature (IUCN) Threats Classification [12]. Outcome: The outcome is reported biodiversity change at genetic, species, community, or ecosystem level. The primary focus is negative change.

Search strategy

The literature search will be conducted on Web of Science Core Collection (WoS). The search string is designed to prioritize recall over precision, capturing a broad corpus (see Table 1). It comprises five components combined using AND: a publication-type filter (A); a scientific disciplines filter via WoS subject categories (B); a change/assessment terms block (C); five driver-specific sub-strings, one per IPBES direct driver category, combined using OR and adopted from a prior rapid evidence assessment [6] (D); and a publication year filter, spanning 2000–2026 (E). We adopt 2000 as the start year of our search because it marks the close of the Convention on Biological Diversity's (CBD) foundational period. In 1993, the CBD entered into force [11]; by March 1998, 86 Parties had submitted national reports on their Article 6 implementation measures [16]; and by 2000, it had been ratified by 176 countries [17]. This period closed just before the CBD's 2002 Strategic Plan introduced explicit, time-bound targets. The year 2000 therefore provides a transparent lower bound aligned with the intended temporal scope of the map.

Bibliographic databases

The search will be executed on the WoS Core Collection. Throughout the search, no language restrictions will be applied. Gray literature is excluded in this systematic map; processing of gray literature at this corpus scale falls outside the scope of the LLM pipeline. This is acknowledged as a limitation of the search coverage.

Web-based search engines

No web-based search engines will be searched.

Organisational websites

No organizational websites will be searched.

Comprehensiveness of the search

Comprehensiveness was assessed against an a priori test list of 581 publications compiled from five published sources (three meta-analyses, a rapid evidence assessment, and a systematic map protocol) on direct drivers of biodiversity loss [5–9] and the IPBES Global Assessment chapters on direct drivers (sections 2.1.12–2.1.17) [18]. Together, these sources span all five IPBES direct driver categories. The test list was assembled by extracting reference lists from each source and excluding gray literature and records not indexed in WoS, yielding 581 records from an initial 791 candidates.

The final search string retrieved 497 of these records, yielding source-specific recall ranging from 71.43% to 97.32% and an overall recall of 85.54%.

Search update

No search updates are planned.

Screening strategy

The literature search will retrieve over 2.4 million bibliographic records spanning all five direct anthropogenic drivers of biodiversity change. All eligibility screening decisions will be made by GPT-5-nano (version 2025-08-07). Prompts and the reasoning effort parameter will be finalized on the training and development splits of the validation datasets and frozen prior to full-corpus application; the prompts-tuning and freezing procedure are described in Supplementary Text S2. The model will apply a four-question procedure (Q1–Q4) to the title and abstract of each record; each question returns a structured label alongside a result of Yes, Unclear, or No. Screening will proceed in three stages: 1. Deduplication: Duplicate records will be removed from the raw retrieved corpus using the WoS primary key of each bibliographic record. 2. Abstract availability: Records without an abstract will be excluded, as the abstract is the primary source of text for screening and data coding. 3. Eligibility screening (Q1–Q4): Each remaining record will be screened using the four-question procedure. A record will be classified as eligible only when all four questions return Yes or Unclear; a single No at any question yields an ineligible classification. Unclear responses will be retained to avoid false exclusions. This map applies no full-text screening stage, differs from CEE guidelines. This enables consistent and reproducible screening at a scale infeasible for traditional systematic maps.

Eligibility criteria

Each of the four screening questions targets one component of the evidence definition (see Figure 1 and Supplementary Text S1). A record will be excluded only when at least one question returns No. 1) Biodiversity change: Does the record report at least one form of biodiversity change? Change is operationalized across four forms: genetic change, species change, community change, and ecosystem change. 2) Direction of change: Does the record identify a direction of biodiversity change? Direction is recorded as negative, positive, mixed, or unclear. The primary analytical focus is negative change, i.e., loss of biodiversity. 3) Direct anthropogenic driver: Does the record mention or plausibly imply at least one direct anthropogenic driver? Drivers are assessed using the five IPBES driver categories [2] and the IUCN Threats Classification Scheme [12]. 4) Linkage: Does the record report or plausibly imply a link between the identified driver and the biodiversity change?

Consistency checking

Eligibility screening differs from CEE guidelines in two respects: all decisions will be made by GPT-5-nano rather than multiple human reviewers, and no full-text screening stage will be applied. The implications of these choices for screening reliability will be assessed through two consistency checks against expert full-text labels, defined as eligibility decisions assigned by the original authors of the reference sources based on full-text reading (see Table 2). Differences in inclusion criteria across reference sources will be considered when interpreting the consistency checks. Consistency Check 1 (CC1) will compare author-assigned labels from abstracts only against expert full-text and LLM labels, establishing a human recall benchmark; Consistency Check 2 (CC2) will evaluate LLM labels against expert full-text labels. As screening will be performed by GPT-5-nano, a single model, conventional inter-rater reliability is not applicable; reliability will be assessed using recall as the primary metric to minimize false exclusions at the screening step. Together, these checks cover approximately 2,900 human-annotated records, drawn from expert full-text sources and author-assigned abstract-level annotations, and constitute the primary human validation of model outputs. Following full-corpus application, a sample of screened records will be reviewed post hoc to assess

the consistency of the resulting decisions. Limitations concerning model transparency and reproducibility are presented in the Study Validity Assessment section.

Reporting screening outcomes

Screening outcomes will be reported in full to support transparency and reproducibility. The following will be provided: 1) A Reporting Standards for Systematic Evidence Syntheses (ROSES) flow diagram documenting record counts at each stage of the screening pipeline, from raw retrieval to final eligible records. 2) The raw literature retrieval dataset exported from WoS in Parquet format. 3) The full screening output dataset in Parquet format. The field structure is defined in Supplementary Table 1. 4) The LLM prompt, model configuration, and raw model output at each eligibility stage for each record.

Study validity assessment

Under CEE guidelines, critical appraisal of study validity is optional for systematic maps and does not influence the mapping process itself. No formal risk-of-bias appraisal will be conducted for exclusion or weighting of included studies. Instead, study validity indicators will be captured through the coding process: study design, data collection method, and analysis method will be coded for every eligible record via the study attributes label set (L5).

Consistency checking

As no formal critical appraisal is conducted, no dedicated consistency checking for this stage is required. We use this section, however, to address human validation and LLM limitations spanning both screening and coding. We follow recent guidance [23] in line with earlier work [24, 25] and perform human validation at two stages. First, prior to full-corpus application, LLM labels will be assessed against more than 8,000 human-annotated coding records and approximately 2,900 human-annotated screening records. Second, following full-corpus application, we will conduct a post-hoc manual review of a sample of labeled records. LLM-based labeling introduces two major limitations. First, screening and coding rely on titles and abstracts rather than full text; broad information such as geographic scope or primary drivers is typically stated, but finer-grained detail such as a threat's specific mechanism may not be, an inherent limitation of abstract-based coding. Second, GPT-5-nano is a closed-weight, proprietary model, which limits independent verification of its training data and internal decision processes. We will report the limitations of the LLM approach using the GUIDE-LLM reporting checklist, a consensus-based reporting framework [26]. The checklist is designed to make potential limitations, hidden biases, and assumptions more explicit, thereby strengthening the rigor and transparency of our research (see Extended Data Table 1 for a completed draft checklist).

Data coding strategy

Eligible records will be coded across six label sets to construct an evidence chain: which drivers are reported as linked to biodiversity change (L1), through which specific threat categories (L2), in which locations and ecosystems (L3; L4), characterized by which study attributes (L5), and affecting which taxa (L6). All coding decisions will be made by GPT-5-nano (version 2025-08-07). The model will code each record by applying pre-defined decision rules to its title and abstract. Prompts will use zero-shot instructions, refined iteratively on training and development splits prior to full-corpus application (see Supplementary Text S2 for the prompts-tuning and freezing procedure). Where applicable, label sets will draw on controlled vocabularies from established classification schemes: the IPBES Direct Drivers, the IUCN Threats Classification Scheme, IPBES Geographic Regions, the Global Ecosystem Typology (GET), and the GBIF Backbone Taxonomy [2, 12–15]. Two coding modes will be applied depending on label set structure. Flat-structured label sets (L1, L3, L5) will be coded in a single prompt call per record. Hierarchical label sets (L2, L4) will be coded in staged multiple-pass sequences, where each level's output constrains candidate labels at the next level, avoiding inconsistent assignments. The taxonomy label set (L6) will be applied conditionally: only records

flagged by L5 as taxon-focused will enter the taxonomy coding step.

Meta-data to be coded

Six label sets will be coded for each eligible record (see output schema in Supplementary Table 3). IPBES Direct Drivers (L1): broad direct anthropogenic driver category as per the IPBES framework [2]. IUCN Threats (L2): specific proximate threats to biodiversity, resolved across three hierarchical levels [12]. Geographic Scope (L3): IPBES region, sub-region, and country [13], together with named sub-national locales and coordinates where reported or inferred from locale names. Ecosystem Typology (L4): Global Ecosystem Typology (GET) realm, biome, and Ecosystem Functional Group [14]. Study Attributes (L5): study design, methods of data collection and analysis, comparator presence and type (where reported), and a taxon-focus flag. Taxonomy (L6): taxonomic rank and scope from kingdom to species, resolved against the GBIF Backbone Taxonomy [15], applied conditionally to records where L5 returns a taxon-focus flag.

Consistency checking

Data coding differs from CEE guidelines in the same two respects as eligibility screening; the implications of these choices for coding reliability will be assessed through two consistency checks against expert full-text labels, defined as coded labels assigned by the original authors of the reference sources based on full-text reading (see Table 3). Consistency Check 1 (CC1) will compare author-assigned labels from abstracts only against expert full-text labels, providing a human-performance benchmark for each label set; Consistency Check 2 (CC2) will evaluate LLM labels against expert full-text labels across all six label sets (L1-L6). Because all coding is performed by a single model, conventional inter-rater reliability is not applicable; reliability will instead be assessed using precision, recall, F1, and Cohen's κ . Together, these checks cover more than 8,000 human-annotated records, drawn from expert full-text sources and author-assigned abstract-level annotations, and constitute the primary human validation of model outputs. Following full-corpus application, a sample of records from each label set will be reviewed post hoc to assess the consistency of the resulting labels. Limitations concerning model transparency and reproducibility are presented in the Study Validity Assessment section.

Type of mapping

This systematic map will be produced as a coded tabular dataset in which each row represents one eligible record enriched with labels from all six coding dimensions (L1-L6). Because the pipeline is fully automated and reproducible, the map is designed as a living systematic map: it can be re-run against new literature without manual screening and coding. The dataset will combine a full bibliographic record export from WoS with the metadata fields defined in Supplementary Table 3. Full screening outputs will be provided as a separate dataset; the field structure is defined in Supplementary Table 1. Both will be made available in a publicly accessible repository alongside the map. The coded dataset will be accompanied by an interactive web application for evidence exploration.

Narrative synthesis methods

Narrative synthesis will characterize the distribution of evidence across all six coded dimensions through descriptive statistics on counts and proportions per category. Cross-tabulations of driver by geography, driver by ecosystem, and driver by study attributes will be reported as matrices. Temporal trends in publication volume by driver category will document how the evidence base has evolved since 2000. No effect sizes are estimated, and no causal conclusions are drawn.

Knowledge gap identification strategy

Knowledge gaps will be identified by inspecting cross-tabulation matrices for sparse or absent driver-dimension combinations across geography, ecosystems, study attributes, and taxa.

Demonstrating procedural independence

Eligibility screening and data coding will be performed by an LLM pipeline that applies fixed prompts and decision criteria uniformly to all records. Any records authored by review team members will be processed by the same frozen LLM pipeline as all other records; no reviewer discretion will be applied. All pipeline code, prompts, and decision rules will be documented in the supplementary code repository.

Competing interests

The authors declare no competing interests.

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No external funding was received.

Author's contributions

H.M. developed the protocol and wrote the manuscript. K.F. reviewed the manuscript and provided feedback. S.F. conceived the study, provided supervisory guidance, and reviewed the manuscript. All authors read and approved the final protocol manuscript.

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